In Excel VBA, the code you write is organized into **Procedures**. Understanding the difference between a **Sub Procedure** (Sub) and a **Function Procedure** (Function) is essential for building structured and efficient automation, especially when managing complex engineering data.

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1. Sub Procedures (Sub)

A **Sub** is a series of VBA statements that performs a specific action but **does not return a value** to the cell or the calling code.

- **Execution**: You can run a Sub by clicking a button, using a keyboard shortcut, or calling it from another procedure.
- **Purpose**: Used for "doing things," such as formatting a "Salary statement," creating **Charts**, or sorting a list of **1,048,576 rows**.
- **Syntax**:
- VBA

```
Sub FormatHeader()
   Range("A1:Z1").Font.Bold = True
   Range("A1:Z1").Interior.Color = vbCyan
End Sub
```

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2. Function Procedures (Function)

A **Function** is a series of statements that performs a calculation and **returns a single value**.

- **Execution**: Functions can be called within other VBA procedures or used directly in an Excel worksheet just like built-in formulas such as **SUMIF** or **VLOOKUP**.
- **Purpose**: Used for "calculating things," such as a custom mathematical model for **Neural Networks** or a specialized tax calculation for a payroll sheet.
- **Syntax**:
- VBA

```
Function CalculateTax(salary As Double) As Double
   CalculateTax = salary * 0.15
End Function
```

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3. Key Differences

Feature	Sub Procedure	Function Procedure
Returns Value	No.	Yes.
Worksheet Use	Cannot be used as a formula.	Can be used as a custom formula.
Primary Goal	To perform actions (e.g., Delete rows).	To return a result (e.g., Calculate average).
Calling	Run via **Alt+F8**, button, or event.	Called by name in a cell or code.

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4. Passing Arguments

Both Subs and Functions can accept "arguments" (inputs) to make them more dynamic.

- **ByVal (By Value)**: Passes a copy of the data; the original variable remains unchanged.
- **ByRef (By Reference)**: Passes a pointer to the original data; if the procedure changes the value, it changes the original variable as well (this is the default in VBA).

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5. Practical Integration

- **Modular Programming**: You can write a **Function** to calculate a specific data point (like a "Total" score) and then call that function within a **Sub** that loops through all your records.
- **User Interaction**: You might use a **Sub** to trigger an **Input Box** that asks for a value, then pass that value to a **Function** for processing.

- **Error Handling**: Functions are excellent for "cleaning" data—for example, a function that checks if a "Student Name" is valid before a **Sub** attempts to save the file.